Circle — Definition, Forms, Reality Check

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

LESSON 01 / 07

Definition: fixed distance from a centre

A circle with centre $C=(h,k)$ and radius $r\ge 0$ is the locus

$$(x-h)^2+(y-k)^2=r^2.$$

For $r>0$ this is a proper circle; $r=0$ is the single point C.

SHORT DERIVATION

Complete squares in the general equation

For $x^2+y^2+2gx+2fy+c=0$, group the coordinate terms:

$$(x+g)^2+(y+f)^2=g^2+f^2-c.$$

Thus the centre is $(-g,-f)$ and $r^2=g^2+f^2-c$. The equal coefficients of x^2, y^2, with no xy term, are essential for a circle in orthogonal Cartesian axes.

Three cases—do not call every equation a circle

- $g^2+f^2-c>0$: a real circle of positive radius.
- $g^2+f^2-c=0$: a point-circle (one real point).
- $g^2+f^2-c<0$: no real points.

WORKED EXAMPLE

Centre, radius, then check

$x^2+y^2-2x-4y-4=0$ gives $g=-1, f=-2, c=-4$, centre $(1,2)$ and $r^2=1+4+4=9$, so $r=3$. Check $(1,5)$: it is distance 3 from $(1,2)$ and satisfies the equation.

COMMON TRAP

Radius is the nonnegative square root

If a formula yields $r^2<0$, it is not a real circle. If $r^2=0$, do not describe it as a circle with positive circumference. A negative value cannot be repaired by dropping a sign.

LESSON 02 / 07

Three independent conditions determine a circle

A general circle has equation $x^2+y^2+Dx+Ey+F=0$ with three unknown coefficients. Substituting three noncollinear points gives a linear system for D, E, F. If the points are collinear, no proper circle passes through all three.

WORKED EXAMPLE

Through the origin and two unit intercepts

Through $(0,0), (1,0), (0,1)$: the origin gives $F=0$, then $1+D=0$ and $1+E=0$. The circle is $x^2+y^2-x-y=0$, centre $(1/2, 1/2)$, radius $1/\sqrt{2}$.

Diameter endpoints give a dot-product equation

For diameter endpoints $A=(x_1,y_1)$ and $B=(x_2,y_2)$, a point $P=(x,y)$ lies on the circle exactly when $\angle APB=90^\circ$ (Thales' theorem). Therefore

$$(P-A)\cdot(P-B)=0,$$

or explicitly

$$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0.$$

The centre is the midpoint of AB and the radius is $AB/2$.

WORKED EXAMPLE

Diameter through two points

For $A=(1,2)$, $B=(5,8)$, $(x-1)(x-5)+(y-2)(y-8)=0$, i.e. $(x-3)^2+(y-5)^2=13$. Both endpoints satisfy the diameter equation.

Parametric equations

For centre (h,k) and radius r,

$$x=h+r\cos t,\qquad y=k+r\sin t,\qquad 0\le t<2\pi.$$

Determine a Circle and Parametrize It (cont.)

The angle parameter traces every point once except the equivalent endpoint at 2π.

A rational parameter is

$$x = h + r\frac{1-u^2}{1+u^2}, \qquad y = k + r\frac{2u}{1+u^2}, \qquad u \in \mathbb{R}.$$

It traces the circle except $(h-r, k)$; that missing point is the limit $|u| \to \infty$. Include it separately when a problem asks for every point.

COMMON TRAP

Parameter is not arc length

In the trigonometric parametrization, t is the polar angle about the centre. In the rational form, $u = \tan(t/2)$ and the omitted point comes from the half-angle coordinate's pole.

LESSON 03 / 07

Centre-to-line distance classifies the intersection

Let a circle have centre $C = (h, k)$, radius r, and let the line be $ax + by + c = 0$. The distance from C to the line is

$$d = \frac{|ah + bk + c|}{\sqrt{a^2 + b^2}}.$$

- $d < r$: two intersection points (a secant).
- $d = r$: one repeated intersection (a tangent).
- $d > r$: no real intersection.

SHORT DERIVATION

Why the chord length is twice a right-triangle leg

The perpendicular from the centre bisects a chord. If its length is ℓ, the right triangle has hypotenuse r and one leg d, hence $(\ell/2)^2 = r^2 - d^2$ and

$$\ell = 2\sqrt{r^2 - d^2}.$$

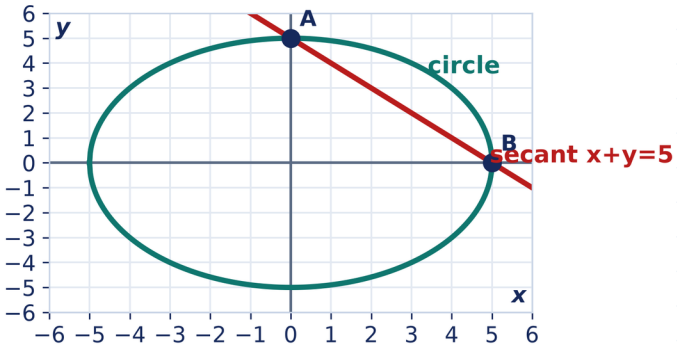
This is real precisely in the secant/tangent cases $d \le r$.

WORKED EXAMPLE

A secant, not merely a line equation

The line $x + y = 5$ is at distance $5/\sqrt{2}$ from the centre $(0, 0)$ of $x^2 + y^2 = 25$. Since this is less than 5, it cuts a chord of length $2\sqrt{25 - 25/2} = 5\sqrt{2}$.

Centre-to-line distance sets the chord length



Solve the intersection only after classifying it

Parametrize the line by $P(t) = P_0 + t(-b, a)$, where P_0 is any point on it. Substitute in $(x-h)^2 + (y-k)^2 = r^2$: a quadratic in t results. Its discriminant is positive, zero or negative exactly when $d < r$, $d = r$ or $d > r$.

COMMON TRAP

Tangency is a repeated root

A tangent line meets the circle in one geometric point but gives a double root after substitution. Do not infer "no intersection" from a zero discriminant.

LESSON 04 / 07

Tangent: perpendicular to the radius

For $(x-h)^2 + (y-k)^2 = r^2$ and a point $T = (x_1, y_1)$ on the circle, the tangent is

$$(x_1 - h)(x - h) + (y_1 - k)(y - k) = r^2.$$

The normal is the line through T and the centre (h, k).

Tangent and Normal at a Point (cont.)

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

SHORT DERIVATION

The tangent equation is a dot-product condition

The radius vector $T - C$ is normal to the tangent. Thus X is on the tangent exactly when $(T - C) \cdot (X - T) = 0$. Rearranging gives $(T - C) \cdot (X - C) = |T - C|^2 = r^2$.

For $x^2 + y^2 + 2gx + 2fy + c = 0$, the same tangent is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

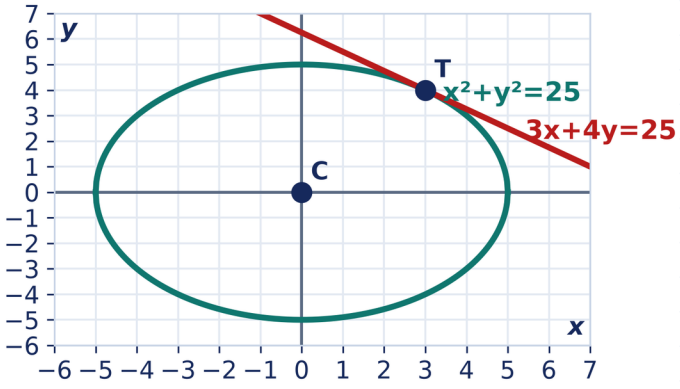
This formula assumes T lies on the circle; the "$T = 0$" replacement is a mnemonic, not a substitute for checking incidence.

WORKED EXAMPLE

Unit-free tangent check

For $x^2 + y^2 = 25$ at $T = (3, 4)$, the tangent is $3x + 4y = 25$. It contains T, and its normal vector $(3, 4)$ is the radius direction.

The radius is normal to the tangent



COMMON TRAP

A tangent is attached to a point on the circle

If a proposed point is not on the circle, the at-point tangent formula does not apply. From an external point there are generally two tangents; derive them as lines through that point with discriminant zero or use the polar/chord-of-contact equation.

LESSON 05 / 07

Power records inside, on, or outside

For circle centre $C = (h, k)$ and radius r, define

$$\text{Pow}(P) = |P - C|^2 - r^2.$$

It is negative inside, zero on, and positive outside. In the general equation $x^2 + y^2 + 2gx + 2fy + c = 0$, the power at (x_0, y_0) is the left-hand side evaluated there.

SHORT DERIVATION

Tangent length from a right triangle

If P is outside and T is a tangent point, $CT \perp PT$. Therefore

$$PT^2 = PC^2 - CT^2 = PC^2 - r^2 = \text{Pow}(P).$$

So each tangent from P has length $\sqrt{\text{Pow}(P)}$.

WORKED EXAMPLE

Tangent length is not centre distance

From $P = (5, 5)$ to $x^2 + y^2 = 9$, power is $25 + 25 - 9 = 41$. Each tangent has length $\sqrt{41}$, whereas $PC = \sqrt{50}$.

Chord of contact / polar of an external point

For a circle centred at the origin, $x^2 + y^2 = r^2$, the two tangent points from $P = (p, q)$ lie on the line

$$px + qy = r^2.$$

This line is the ***chord of contact***. Its intersection with the circle gives the two contact points. For a translated circle, first use coordinates relative to its centre: $(P - C) \cdot (X - C) = r^2$.

WORKED EXAMPLE

Find both contacts by one line and one circle

From $P = (5, 0)$ to $x^2 + y^2 = 9$, the chord of contact is $5x = 9$, so $x = 9/5$. Then $y^2 = 9 - 81/25 = 144/25$, giving contacts $(9/5, \pm 12/5)$. The tangent length is $\sqrt{25 - 9} = 4$.

COMMON TRAP

The polar formula needs the same circle centre

For a circle not centred at the origin, replace P and X by their vectors from the centre. Forgetting this translation shifts the chord of contact.

LESSON 06 / 07

Subtract normalized circle equations

Write $S_i = x^2 + y^2 + D_i x + E_i y + F_i$ for $i = 1, 2$; the corresponding circles are $S_i = 0$. Their power functions are $S_1(x, y)$ and $S_2(x, y)$. Equal power means $S_1 - S_2 = 0$; the quadratic terms cancel, leaving a line (unless the circles are concentric or identical).

SHORT DERIVATION

Why the line is perpendicular to the centres

The centre difference is parallel to $(D_1 - D_2, E_1 - E_2)$, the normal vector of $S_1 - S_2 = 0$. Thus the radical axis is perpendicular to the line of centres. When the real circles intersect, its points are exactly their common chord.

WORKED EXAMPLE

Subtract, then test equal powers

For $x^2 + y^2 = 9$ and $x^2 + y^2 - 4x + 6y - 3 = 0$, subtract to get $2x - 3y - 3 = 0$. At $(0, -1)$ both power values equal -8, checking the line's equal-power meaning.

Three circles and the radical centre

The radical axes of three circles are the pairwise equal-power lines. If the circles are not identical and the axes are not parallel, solve two axes to get their common point; equality of the third pair follows by subtraction. Degenerate cases need care: concentric unequal circles have no finite equal-power locus, identical circles have equal power everywhere, and parallel axes need not meet in a finite radical centre.

COMMON TRAP

A radical axis need not be a real common chord

Two nonconcentric circles can be disjoint or nested. Their equal-power line still exists, but it contains real points only as a locus of equal powers—not points on both circles. Call it a common chord only when the circles actually intersect.

LESSON 07 / 07

Compare centre distance and radii

For centres distance d and radii $r_1, r_2 > 0$:

Condition	Real intersection
$d > r_1 + r_2$	none; externally separate
$d = r_1 + r_2$	one; externally tangent
$\lvert r_1 - r_2 \rvert < d < r_1 + r_2$	two points
$d = \lvert r_1 - r_2 \rvert > 0$	one; internally tangent
$0 < d < \lvert r_1 - r_2 \rvert$	none; one strictly inside the other
$d = 0$	concentric: compare radii; identical if radii agree

The zero-radius point-circle is handled separately by testing whether its point lies on the other circle.

Efficient order of attack

Efficient order of attack

1. Complete squares and identify centres/radii.
2. For a line, use centre-to-line distance before solving intersections.
3. For a tangent, check the point lies on the circle and use the radius as normal.
4. For a common chord, subtract normalized equations first.
5. For two circles, compare d with $r_1 + r_2$ and $|r_1 - r_2|$.

WORKED EXAMPLE

Two equal circles

$(x-0)^2 + y^2 = 4$ and $(x-3)^2 + y^2 = 4$ have $d = 3$, $r_1 + r_2 = 4$, and $|r_1 - r_2| = 0$. Hence they meet twice. Subtracting gives radical axis $x = 3/2$; substituting yields $y = \pm\sqrt{7}/2$.

ASSUMPTIONS & CONVENTIONS

JEE boundary

The official circle bullet includes circle forms, tangent, normal, chord, parametric equations and intersection with a straight line. Radical axes, power and two-circle position are useful direct applications; the official scope is not a promise to include every theorem about circles.

Coaxal Families and Radical Centre

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

LESSON 01 / 03

A normalized pencil of circles

Write $S_i = x^2 + y^2 + D_i x + E_i y + F_i$ for $i = 1, 2$, normalizing each circle to the same $x^2 + y^2$ coefficient. Then

$$S_\lambda = (1 - \lambda)S_1 + \lambda S_2 = 0$$

is a family of circles. Every member has the same radical axis $S_1 - S_2 = 0$, and if the original circles intersect in two real points, every member passes through both. If they do not meet in real points, the radical axis is still an equal-power line but is not a real common chord.

WORKED EXAMPLE

A pencil through two fixed points

The family $x^2 + y^2 + \lambda(x - y) - 2 = 0$ passes through $(1, 1)$ and $(-1, -1)$ for every real λ. Subtracting two members gives $(\lambda_1 - \lambda_2)(x - y) = 0$, so their radical axis is $x - y = 0$.

Radical centre, when finite

For three circles, solve two radical-axis equations. If the axes are nonparallel, their intersection is the unique finite radical centre, and equal-power transitivity makes powers to all three circles equal. If axes are parallel or coincide, do not force a finite point.

COMMON TRAP

Concentric degeneracy from the source

For $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$, equal powers would require $-1 = -4$, impossible: these concentric unequal circles have no finite radical axis. A third circle cannot create a finite point equidistant in power from both. Replace one circle with a nonconcentric one before asking for a radical centre.

LESSON 02 / 03

Inversion in a circle

Inversion of radius ρ about centre O sends $P \neq O$ to P' on ray OP so that $OP \cdot OP' = \rho^2$. With O at the origin,

$$P' = \frac{\rho^2 P}{|P|^2}.$$

It is involutive and reverses orientation while preserving angles in magnitude. The centre has no finite image.

SHORT DERIVATION

A line not through the inversion centre becomes a circle

For unit inversion, write $x = X/(X^2 + Y^2)$ and $y = Y/(X^2 + Y^2)$. The line $x = a \neq 0$ becomes $X = a(X^2 + Y^2)$, or $(X - 1/(2a))^2 + Y^2 = 1/(4a^2)$: a circle through the origin. A line through the origin stays a line; conversely, a circle through the origin maps to a line not through it.

Distance ratio is a circle, with a line as the limiting case

For distinct A, B, the locus $PA/PB = k$ is the perpendicular bisector when $k = 1$ and a circle when $k \neq 1$. The centre lies on AB and the circle degenerates to A at $k = 0$. This is the Apollonius locus; a derivation and coordinate-safe formula are in Chapter 12.

COMMON TRAP

Check for the exceptional ratio

Do not divide by $1 - k^2$ before splitting off $k = 1$. For equal distances the locus is a line, not a circle of "infinite radius" in an ordinary coordinate calculation.

LESSON 03 / 03

Simson Line and Ptolemy

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

Simson line: an exact hypothesis

Let ABC be a nondegenerate triangle and P any point. The feet of perpendiculars from P to the three **lines** BC, CA, AB are collinear if and only if P lies on the circumcircle of ABC. The word "lines" matters: a foot may fall on an extension of a side.

SHORT DERIVATION

The cyclic quadrilaterals behind Simson

Call the feet $D \in BC$, $E \in CA$, $F \in AB$. The right angles show P, D, C, E and P, D, B, F are cyclic. Directed angles modulo 180° give

$$\angle EDF = \angle EDP + \angle PDF.$$

The two right-angle circles give $\angle EDP = \angle ECP$ and $\angle PDF = \angle PBF$. Since E, C, A and F, B, A lie on the same side-lines,

$$\angle ECP = \angle ACP, \qquad \angle PBF = \angle PBA.$$

If A, B, C, P are concyclic, $\angle ACP = \angle ABP$, while $\angle PBA = -\angle ABP$ as directed angles, so $\angle EDF = 0$ and the feet are collinear. Reversing the same equalities proves the converse in the nondegenerate case. Numerical collinearity alone is evidence, not a proof.

Ptolemy: order the vertices around the circle

For a convex cyclic quadrilateral A, B, C, D in circular order,

$$AC \cdot BD = AB \cdot CD + BC \cdot DA.$$

SHORT DERIVATION

A similarity proof

Choose E on diagonal AC so that $\angle ABE = \angle CBD$. Since A, B, C, D are cyclic and E lies on AC, triangles ABE and CBD are similar, giving $AE \cdot BD = AB \cdot CD$. The remaining triangles BCE and BDA are similar, giving $CE \cdot BD = BC \cdot DA$. Add and use $AE + CE = AC$ to obtain Ptolemy.

WORKED EXAMPLE

Pythagoras is the rectangle case

For a rectangle with adjacent sides a, b and both diagonals c, Ptolemy gives $c^2 = a^2 + b^2$. The rectangle condition matters: a general cyclic quadrilateral with one diagonal a diameter does not by itself make Ptolemy reduce directly to that equation.

COMMON TRAP

Source-paper consistency correction

A cyclic quadrilateral with successive sides $3, 4, 5, 6$ and a diagonal of length 7 cannot exist: that diagonal and the adjacent sides $3, 4$ would form a degenerate triangle ($3 + 4 = 7$). Ptolemy's numerical equation alone is not an existence proof for arbitrary proposed side/diagonal data.